Airbnb Price Prediction

Model Evaluation and Results

1. Problem Statement

This project investigates whether machine learning models can estimate Airbnb listing prices from listing characteristics and location information across six US cities.

2. Dataset and Methodology

The dataset contains 74,111 listing records. The target variable is log_price. The project compares Linear Regression, Random Forest, and XGBoost using MAE, RMSE, and R^2 on a held-out test set.

3. Model Performance

Model	MAE	RMSE	R^2
Linear Regression	0.3056	0.4187	0.6588
Random Forest	0.2830	0.3952	0.6960
XGBoost	0.2843	0.3911	0.7022

MAE and RMSE are reported on the log-price scale; they are not direct dollar-error measurements.

4. Results and Interpretation

XGBoost achieved the highest R^2 (0.7022) and lowest RMSE (0.3911). Random Forest achieved the lowest MAE (0.2830). XGBoost therefore performed best on two of the three reported metrics, but it did not achieve the lowest error on every metric.

5. Limitations

The dataset covers six US cities and may not generalise to other markets. Prediction quality depends on the available features, data quality, and factors not represented in the dataset. Feature importance describes predictive contribution and does not prove causation.

6. Conclusion

The experiments show that the evaluated regression models capture useful patterns in Airbnb listing prices. XGBoost offers the strongest overall performance by R^2 and RMSE, while Random Forest has a slightly lower MAE. Predictions should be treated as estimates rather than guaranteed listing prices.